

Masoud Kargar

AI Engineer | AI Backend Engineer | Senior .NET Backend Developer

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PROFESSIONAL SUMMARY

AI Engineer / AI Backend Engineer with a strong .NET backend foundation and 6+ years of software engineering experience. Hands-on experience building Retrieval-Augmented Generation systems, integrating LLM APIs, embeddings, vector databases and semantic search into production-oriented backend architectures. Built an OpenAI-compatible RAG API with ChromaDB, OpenRouter/OpenAI-compatible providers and a separate FastAPI local Persian embedding service. Strong background in scalable APIs, microservices, authentication, SQL optimization and AI service integration.

AI ENGINEERING

- LLM Application Development, Generative AI, Prompt Engineering, System Prompts
- Retrieval-Augmented Generation (RAG), document ingestion, chunking/retrieval concepts, context augmentation
- Embeddings, Persian/Farsi embeddings, vector similarity, semantic search, ChromaDB
- OpenAI-compatible APIs, OpenAI, OpenRouter, streaming/non-streaming generation
- AI API design, provider abstraction, local embedding microservices
- Python, FastAPI, Uvicorn, Docker, backend integration and deployment

BACKEND ENGINEERING

- C#, .NET 8, ASP.NET Core, REST APIs, Clean Architecture, CQRS, Microservices
- SQL Server, PostgreSQL, MongoDB, Redis, Entity Framework, Dapper
- RabbitMQ, Kafka, Keycloak, JWT, OAuth2/OIDC, Docker, Kubernetes, Linux

PROFESSIONAL EXPERIENCE

Senior .NET Backend Developer | *Rasta Novin Aria* | *Tehran, Iran* | *Apr 2025 – May 2026*

- Designed and developed backend services and RESTful APIs using .NET 8 as part of the core engineering team.
- Improved production SQL query and API performance by approximately 35% through targeted query optimization and architectural refinements.
- Implemented distributed authentication and authorization using Keycloak and JWT across services.

Senior Backend Developer | *Setigh Fara Ofogh* | *Tehran, Iran* | *Oct 2024 – Mar 2025*

- Applied Clean Architecture principles to improve maintainability, testability, and long-term scalability.
- Led migration of a monolithic application toward a microservices-based architecture, improving deployment flexibility and resilience.
- Implemented Keycloak-based authentication for secure inter-service and user-facing communication.

Senior Backend Developer — Freelance | *Self-Employed* | *Pakdasht, Iran* | *Jul 2023 – Oct 2024*

- Built multi-tenant APIs enabling multiple client organizations to operate securely on shared infrastructure.
- Developed blockchain-based applications for client projects using Web3 technologies.
- Designed and implemented scalable backend systems following modern architecture and engineering practices.

.NET Backend Developer | *Behineh Kavan Keyfiat* | *Tehran, Iran* | *Jul 2021 – Jul 2023*

- Contributed to the design, development, and maintenance of backend services and RESTful APIs using .NET and SQL Server.

Back-End Developer — Freelance | *Self-Employed* | *Tehran, Iran* | *Mar 2020 – Jul 2021*

- Delivered backend development services for client projects using .NET and SQL Server.

SELECTED AI PROJECT

RAG API — Retrieval-Augmented Generation System | *GitHub*: <https://github.com/MassoudKargar/RAG>

- Built an OpenAI-compatible RAG API integrating document embeddings, ChromaDB vector storage, semantic search, and LLM chat generation.
- Implemented OpenRouter and OpenAI-compatible chat providers with independent embedding-provider abstraction.
- Built a separate local embedding microservice using FastAPI/Uvicorn and xmanii/maux-gte-persian for Persian-language embeddings.
- Implemented collection initialization, document ingestion, semantic search, and streaming/non-streaming chat responses.

EDUCATION

- Diploma

CERTIFICATIONS & TRAINING

- Machine Learning with Python — Maktabkhooneh (2026)
- Microservice Architecture — Nikamooz, 70 hours (2023)
- PRO .NET Ecosystem — Nikamooz, 110 hours (2022)
- ASP.NET MVC — Cando Academy (2019)
- Programming Fundamentals — Cando Academy (2019)

LANGUAGES

- Persian — Native | English — Upper Intermediate

ADDITIONAL

- Military Service — Completed